

CORE QUALITY CONCEPT

User Need and Expectation x Voice of Customer x QFD x CTQ x Kano Model x Root Cause Analysis x Shainin Red X x Taguchi Robust Design x DOE x Controlled Validation x PDCA x Kaizen

SECTION 01 CASE OBJECTIVE

The objective of this case study is to explore how Quality Engineering principles can be applied to improve personalised content discovery in a large scale digital streaming environment. The focus is not simply on increasing the amount of content consumed. The objective is to understand whether users can discover relevant content efficiently while the system continuously adapts to changing interests, viewing behaviour and expectations. The proposed framework treats recommendation quality as an engineering quality problem that can be understood through user requirements, measurable quality characteristics, root cause analysis, robust experimentation and continuous improvement.

SECTION 02 QUALITY PROBLEM

In a large content environment, users may encounter recommendations that do not match their current interests, expectations or viewing context. User interests can change over time, while different users may interpret relevance differently. A Quality Engineering challenge is therefore to understand what users actually need, convert those needs into measurable quality requirements and determine why recommendation quality may not consistently meet those requirements. The objective is to reduce irrelevant discovery experiences while improving the reliability and usefulness of personalised content discovery.

SECTION 03 USER NEED AND EXPECTATION

The first step is to understand the actual user requirement rather than immediately changing the recommendation system. The framework begins by identifying what quality means from the user perspective. Relevant questions include whether the user can find suitable content efficiently, whether recommendations reflect current interests and whether the discovery experience requires unnecessary effort. This creates the foundation for defining measurable quality requirements before technical improvement begins.

USER NEED

→ EXPECTATION

→ QUALITY REQUIREMENT

→ MEASURABLE OUTCOME

SECTION 04 VOICE OF CUSTOMER AND BEHAVIOURAL SIGNALS

User expectations should not be understood through one source alone. The proposed approach combines direct user feedback with behavioural quality signals such as viewing behaviour, search patterns, skipping behaviour, content completion and other relevant interaction signals. Combining these signals provides a broader understanding of what users find relevant, where friction occurs and how expectations may change over time. The purpose is not simply to collect more data. The purpose is to convert user signals into meaningful quality information.

SECTION 05 QFD AND CRITICAL TO QUALITY

Once user requirements are understood, **Quality Function Deployment** can be used to translate those requirements into measurable **Critical to Quality** characteristics. Not every requirement can be improved simultaneously. QFD helps connect what the user expects with measurable quality characteristics and improvement priorities. The focus is therefore on identifying which quality characteristics have the greatest importance and impact from the user perspective.

USER REQUIREMENTS

→ QFD

→ CTQ

→ PRIORITISATION

SECTION 06 KANO MODEL AND USER EXPECTATIONS

Different user requirements do not influence satisfaction in the same way. The Kano Model can be used to classify requirements into Must Be, Performance and Attractive categories. Must Be requirements represent fundamental expectations. Performance requirements influence satisfaction according to their level of performance. Attractive requirements can create additional satisfaction when present even when users may not explicitly expect them. This classification helps ensure that improvement priorities reflect the nature of the user requirement rather than treating every request equally.

MUST BE

Fundamental baseline expectations. Essential discovery stability and smooth navigation without system friction.

PERFORMANCE

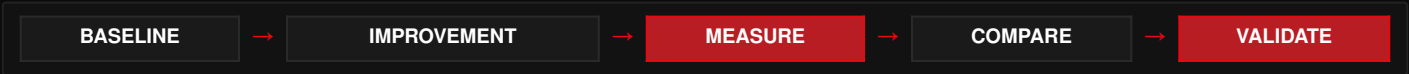
Directly drives satisfaction based on precision. Highly accurate recommendations matching explicit user tastes.

ATTRACTIVE

Delight features creating high satisfaction. Serendipitous content discovery that unexpectedly aligns with user moods.

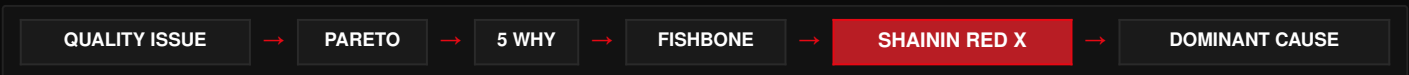
SECTION 07 MEASURING RECOMMENDATION QUALITY

After an improvement is introduced, recommendation quality must be evaluated through measurable outcomes. Relevant quality indicators can include relevant content discovery, recommendation acceptance, unnecessary skips, user satisfaction and decision effort. Before and after comparison can help determine whether the proposed improvement produces a meaningful quality change. The objective is to measure the actual user outcome rather than assuming that a technical algorithmic change automatically represents quality improvement.



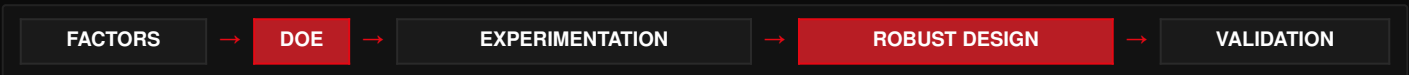
SECTION 08 ROOT CAUSE AND DOMINANT FACTOR ANALYSIS

If the measured outcome shows that recommendations remain problematic, the next step is Root Cause Analysis rather than immediately changing the system. Pareto Analysis can be used to identify the major categories of recommendation quality problems. 5 Why and Fishbone Analysis can then be used to investigate potential causes. **Shainin Red X** can support the identification of the dominant factor contributing to the observed quality problem. The objective is to move from symptom identification to evidence based understanding of the dominant cause.



SECTION 09 TAGUCHI ROBUST DESIGN AND DOE

Once the dominant cause is understood, the improvement should be designed systematically. Taguchi Robust Design and Design of Experiments can be used to evaluate relevant factors and their combinations under different conditions. The objective is not simply to create a solution that performs well for one user group or one situation. The objective is to identify a robust solution that remains effective when relevant user behaviour and operating conditions vary. This approach moves recommendation improvement from simple optimisation toward robust Quality Engineering.



SECTION 10 CONTROLLED A B TESTING AND PILOT VALIDATION

A proposed improvement should not immediately be treated as a final solution. Controlled A B testing or limited pilot validation can be used to compare the proposed experience with the established baseline. Different user behaviours and relevant usage conditions should be considered during validation. If unexpected behaviour, quality degradation or unintended effects appear, the findings should return to Root Cause Analysis and risk evaluation. The solution can then be restructured and tested again before broader implementation.

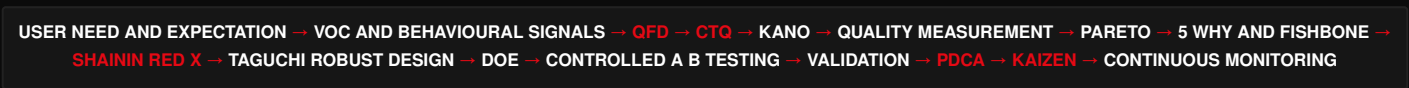


SECTION 11 PDCA, KAIZEN AND CONTINUOUS MONITORING

Quality improvement does not end after a successful experiment. PDCA provides the structure for continuous improvement. Plan defines the quality problem, baseline and improvement objective. Do implements the proposed improvement under controlled conditions. Check evaluates performance, variation, user response and unintended effects. Act standardises the validated improvement or restructures the approach when the expected outcome is not achieved. Kaizen keeps the system open to continuous small improvements as user behaviour and expectations evolve. Continuous monitoring allows emerging quality signals and degradation patterns to become inputs for the next improvement cycle.



SECTION 12 FINAL QUALITY ENGINEERING FRAMEWORK



KEY QUALITY INSIGHT

Recommendation quality is not achieved by producing a relevant suggestion once. It is achieved when user requirements are understood, translated into measurable quality characteristics, classified according to user expectations, analysed through evidence, improved through robust experimentation, validated under variation and continuously refined through feedback and learning.

METHODOLOGY APPLIED:

Voice of Customer • Quality Function Deployment • Critical to Quality • Kano Model • Pareto Analysis • 5 Why Analysis • Fishbone Analysis • Shainin Red X • Taguchi Robust Design • Design of Experiments • Controlled A B Testing • PDCA • Kaizen • Continuous Monitoring

DISCLAIMER: This document is an independent professional research exercise by Jagriti Singh. It is a conceptual Quality Engineering framework developed for academic and professional learning purposes using publicly understood Quality Management and Quality Engineering methodologies. It is not an official Netflix project, internal Netflix analysis, internal Netflix specification, audit, operational document, endorsement, partnership or affiliation.